

Hey, everybody. Welcome to your chapter 5 review. In this episode, it's time to start tackling the different systems of the body. Today we'll cover the nervous, muscular, and skeletal systems. Just like a mechanic needs to learn the individual systems of a car, so too must fitness professionals learn about the systems within the human body. It's important to understand how exercise impacts and influences changes to health and fitness levels. We'll cover these topics today, as well as the learning objectives of the chapter.

Working knowledge of the structure and function of the nervous, muscular, and skeletal systems is foundational. This information is necessary for accurate client assessment, safe exercise, program development, and demonstrating your expertise when answering client questions. Ideally, you should know what comprises each system of the human body and key elements associated with each. To do that, this chapter is really just a launching pad. Your curiosity and commitment to studying these topics beyond this chapter is highly recommended to further develop your knowledge and understanding.

At the drop of a hat, you should be able to explain the role of the nervous system, as well as differences between the central nervous system and the peripheral nervous system, including the different types of sensory receptors and how they relate to exercise programming. When it comes to motor learning, make sure you know how clients learn new movement skills as well, and how the nervous system evolves and changes over time.

Then we move on to the skeletal system. Do you really know its role in the human body? What are its two divisions? How does the skeletal system influence movement?

For the muscular system, what are the specific structures and components that make up muscle tissue? If I say sliding filament theory and excitation contraction coupling, do you know exactly what I mean? Well, if you didn't before, you're about to find out.

Here are the key learning points that you need to know. You should be able to outline the basic structure and function of the nervous system, classify bone and joint types and their associated functions, summarize the structure and function of the muscular system, and identify the effects exercise has on each component of the human movement system. Now that you have your high level overview, it's time to dig a little deeper.

The human movement system includes all the interactions of the nervous, muscular, and skeletal systems. We'll start with the nervous system. It's made up of a specialized network of nerves that transmit information within the human body. And it has many functions, such as providing sensory information to our brain, stimulating human movement through muscle contractions, and keeping the heart and organs functioning. Essentially, it's the wiring and operating system that runs the human machine.

Billions of neurons, or specialized cells, make up the nervous system and process and transmit information through electrical and chemical signals. The nervous system is composed of two interdependent divisions, the central nervous system and the peripheral nervous system.

The central nervous system, or CNS, consists of the brain and spinal cord, and its primary function is to coordinate the activity of all the parts of the body. The peripheral nervous system, or PNS, consists of nerves that connect the CNS to the rest of the body and the external environment. The PNS uses sensory receptors to send information to the CNS.

Sensory receptors are characterized by their response type. Mechanoreceptors respond to mechanical forces, nociceptors respond to pain, chemoreceptors respond to chemical interaction, and photoreceptors respond to light. The muscle spindle and Golgi tendon organ, or GTO, are two important mechanoreceptors to remember.

Muscle spindles live within muscles but run parallel to the muscle fiber. They are sensitive to change in muscle length and the rate of length change. The GTOs are located at the point where skeletal muscle fibers insert into the tendons of skeletal muscle, and they are sensitive to changes in muscular tension and rate of tension change.

The nervous system develops continuously from birth through older adulthood, and plays a major role in how humans learn and retain new skills. Motor skill development, for example, is an interesting area of study that is best understood as a three stage process, cognitive, when a skill is just being learned, associative, when the skill is beginning to be understood, and autonomous, when the skill has been mastered.

Another interesting fact is that the nervous system requires different electrolytes for proper function, which includes sodium, potassium, magnesium, and water. As a fitness professional, it's important to understand sensory and motor information and how that relates to exercise programming. During exercise, the sympathetic nervous system works to increase neural activity and signals different endocrine organs to release hormones that increase heart rate breathing, and alertness.

One example is adrenaline. This is often called fight or flight, where the body is put into a more heightened state and anticipating the need to move. The parasympathetic nervous system has the opposite effect and works to decrease neural activity by suspending the release of fight or flight hormones. This puts the body in a more relaxed state, which is often termed rest and digest. Both systems work together to balance bodily functions and manage interactions with the surrounding world.

Another cool thing about the nervous system as it relates to fitness is how we grow and learn. Over the course of the entire human lifespan, the CNS continues to develop through neuroplasticity and changes in neurocircuitry, those neuronal connections that occur with the acquisition of new skills. For example, when a child is taught how to kick a ball, their brain will stimulate specific neuronal changes and connections that allow them to understand and replicate the newly acquired skill. This concept is highly important for fitness pros, because it mirrors what happens in the brain when a client becomes more skilled at an exercise and is ready to progress to more difficult challenges in the gym.

And now for the system that holds it all together, the skeletal system. It serves many important functions, like providing support for the body and protecting internal organs. The skeletal system has two divisions, axial and appendicular. The axial skeleton is made up of the skull, rib cage, and the vertebral column, and the appendicular skeleton is made up of the arms, legs, and pelvic girdle. Human bones act as attachment sites and levers, or rigid rods, to produce movement when muscles contract, and also to provide support, which translates into posture that is necessary for the efficient distribution of forces acting on the body.

There are five categories of bones, long, short, flat, irregular, and sesamoid. You should be able to identify examples of each of them.

The vertebral column, made up of irregular bones, has five distinct regions, cervical, thoracic, and lumbar, sacrum, and coccyx. In between each vertebra is a disk that acts as a shock absorber and assist with movement.

Bone growth occurs throughout life and remodels itself with specialized cells called osteoclasts, which break down old bone tissue, and osteoblasts, that lay down new tissue. Exercise literally stimulates bone development. You can help a client maintain or improve bone mass through physical activity, specifically resistance training and weight bearing exercise.

The skeletal system also includes our joints. These are formed by one bone articulating with another, and can be categorized by their shape, structure, and function.

In addition to bones, ligaments and their anatomy and purpose in relation to musculoskeletal functioning may be of particular interest to you as a CPT. Ligaments are fibrous tissues that connect bone to bone and provide stability, as well as sensory input to the nervous system that aids proprioception. It's that sixth sense that makes us subconsciously aware of the positioning and movement of our bodies.

It's important to know how movement of skeletal components is classified. Osteokinematics describes bone movement, and arthrokinematics describes movement specifically between joint surfaces. Additionally, synovial joints are worth mentioning, specifically because while they are the most prevalent and have the greatest capacity for movement in the human body. They are also unique, with a synovial capsule, membrane, and cartilage for some padding at the ends of the bones that comprise them, functioning much like engine oil to lubricate the joint, provide nutrition to joint structures, and protect against excessive wear and tear. The membrane secretes synovial fluid within the joint capsule. There are gliding, or plane, condyloid, hinge, saddle, pivot, and ball and socket synovial joints.

Our skeletal system goes through many changes over the course of a lifetime. In addition to exercise impacting bone mass, proper nutrition can also play a role, particularly for older individuals who are prone to osteoporosis that severely decreases bone mass to the point of fracturing. Fitness goals for older adults should focus on improving bone strength and reducing the risk of falls.

OK, that's a lot to take in. Let's move on to muscles.

The three types of muscle in the body are cardiac, smooth, and skeletal. Cardiac muscle is what makes up the heart, and smooth muscle primarily makes up the tissues of internal organs. While each type is vital for the functioning of life itself, skeletal muscle is the one we'll focus on now.

The muscular system links the nervous and skeletal systems, and is responsible for generating the forces that move the human body. Muscles contract to create internal tension that, under the control of the nervous system, manipulates the bones to produce movements around the joints.

Skeletal muscle is made up of individual muscle fibers held together by connective tissues. Their primary functions are to contract and produce movement, support the skeletal system, and assist with homeostasis of the body by producing heat.

Muscles have a complex structure that includes different layers of connective tissue, fascia, epimysium, fascicles, perimysium, and endomysium. These connective tissues play a vital role in movement and allow the forces generated by the muscle to be transmitted to the bones, creating motion. They also allow muscles to work together in functional groups, such as the quadriceps, working together to extend the knee.

Muscles are made up of fascicles, or bundles of muscle fibers, and the muscle fibers are made up of myofibrils. Within each myofibril is a series of sarcomeres. A sarcomere is the functional unit of the muscular system, meaning it is the specific physical site where the muscle contraction occurs.

Sarcomeres contain myofilaments called actin and myosin. These filaments slide past one another to produce a muscle contraction, which involves shortening of sarcomeres. As the sarcomeres shorten, so too do the myofibrils they make up, and collectively this causes the muscle fibers, fascicles, and subsequently the muscle as a whole to shorten. This process is otherwise known as the sliding filament theory.

Adenosine triphosphate, more commonly known as ATP, is needed to create energy for this process. Skeletal muscles won't contract unless they're stimulated by motor neurons. Neural activation represents the communication link between the nervous system and the muscular system.

The nervous system communicates with muscle fibers through a specialized site called the neuromuscular junction, which is a small gap known as a synapse, located between the motor neuron and muscle cells. One motor neuron and the muscle fibers it connects to are known as a motor unit.

Now on to a physiological process called excitation contraction coupling. It sounds pretty exciting, and actually is, in a physiology kind of way. It describes the steps in the muscle contraction process involving the nervous and muscular systems. The electrolyte, calcium, and neurotransmitter acetylcholine are important contributors in the process.

If a nerve impulse is strong enough to trigger an action potential, then it will spread through the whole length of the muscle fiber. If the stimulus isn't strong enough, then there is no muscle contraction. This is known as the all or nothing principle.

Skeletal muscles have some classifications, as well. But don't worry, there are only a couple to remember here. Type 1, or what we call slow twitch muscle fibers, are smaller in size, produce less force, and are fatigue resistant. They contract slowly and generate relatively small forces. This is particularly important for activities that require sustained muscle contractions, like the maintenance of correct posture.

Type 2, or fast twitch muscle fibers, are larger in size, produce more force, and fatigue quicker than type 1 muscles. This is especially important for physical activities that require large forces, like running or jumping. All muscles have varying combinations of both type 1 and type 2 muscle fibers.

Keep in mind that it's ligaments that connect bone to bone, but you also have tendons that connect muscles to bones. Commonly discussed tendons are the Achilles tendon at the ankle and the patellar tendon of the knee. When a tendon is overstretched or torn, this is known as a strain. A ligament that's overstretched or torn is known as a sprain.

The human movement system develops as humans age, from childhood to adulthood. Each system has its own set of normal parameters for growth, which are helpful to learn, while still keeping in mind that a lot of factors can affect growth rates. Exercise and proper nutrition can have a major positive impact on every system, but especially when it comes to bone mass with aging adults.

When you think about it, everything has a growth process. We learn to walk as a baby and learn new moves in the gym. It's a lifelong education, but they are more closely related than you might think, considering how the nervous, skeletal, and muscular systems operate and work together.

Now that you have some basic understanding, I hope it inspires you to incorporate what you've learned into your program designs and client engagements. In the meantime, stay well and happy studying!